Sinusoidal Steady-State Analysis

Video 13

Electrical Science

Dr. Mohamed Bingabr
University of Central Oklahoma

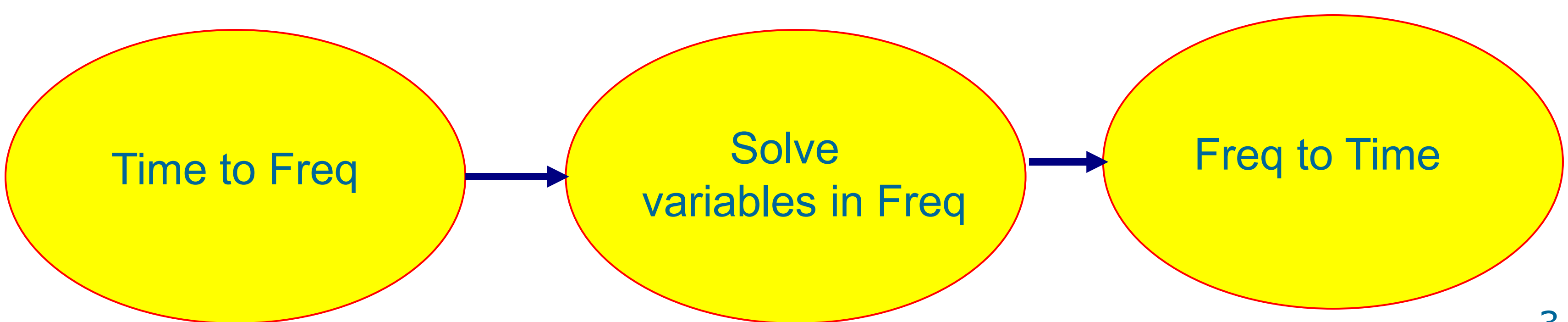
Sinusoidal Steady-State Analysis

- Basic Approach
- Nodal Analysis
- Mesh Analysis
- Superposition Theorem
- Source Transformation
- Thevenin and Norton Equivalent Circuits
- Frequency Response $H(\omega)$

Basic Approach

Steps to Analyze AC Circuits:

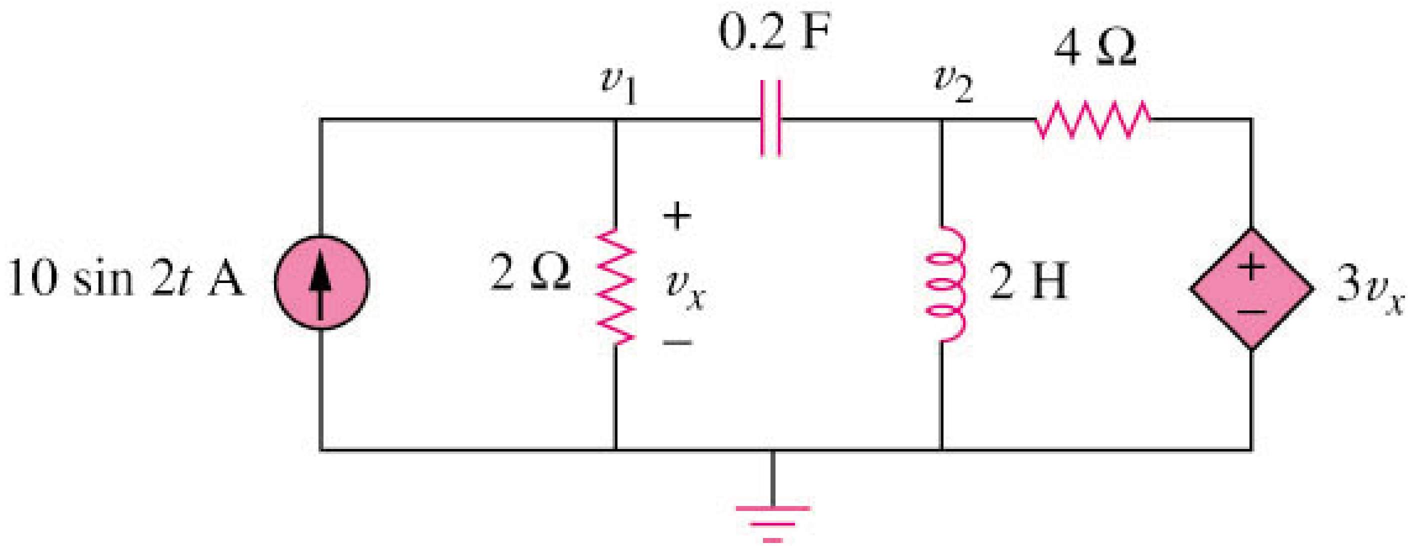
1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.



Nodal Analysis

Example 1

Using nodal analysis, find v_1 and v_2 in the circuit of figure below.



Answer:

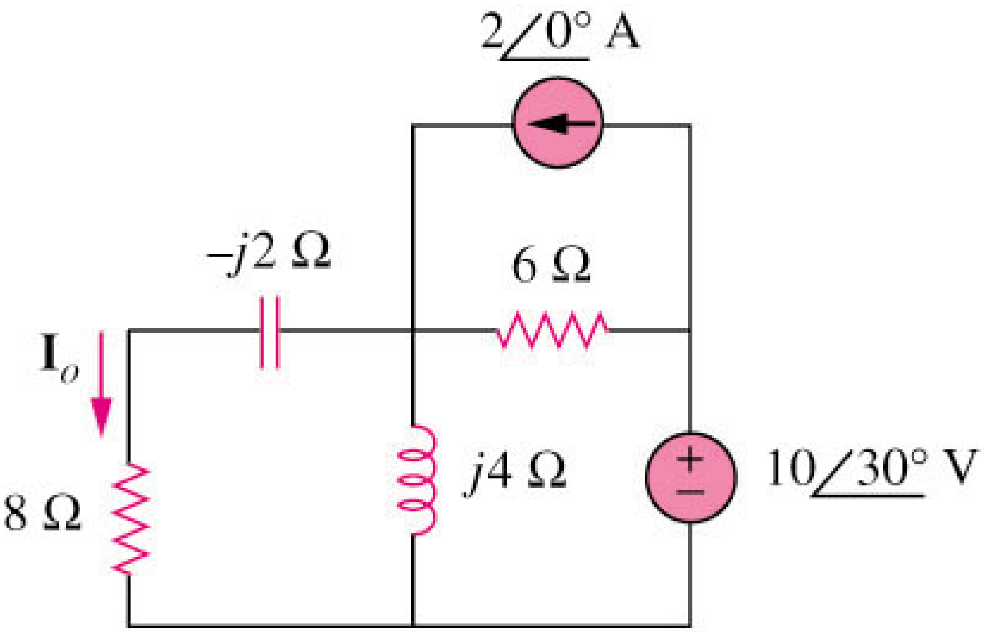
$v_1(t)$ = **11.32 sin(2t + 60.01°) V**

v_2(t) = 33.02 sin(2t + 57.12°) V

Mesh Analysis

Example 2

Find I_o in the following figure using mesh analysis.



Answer: $\mathbf{I_o = 1.194 \angle 65.44^o\ A}$

Superposition Theorem

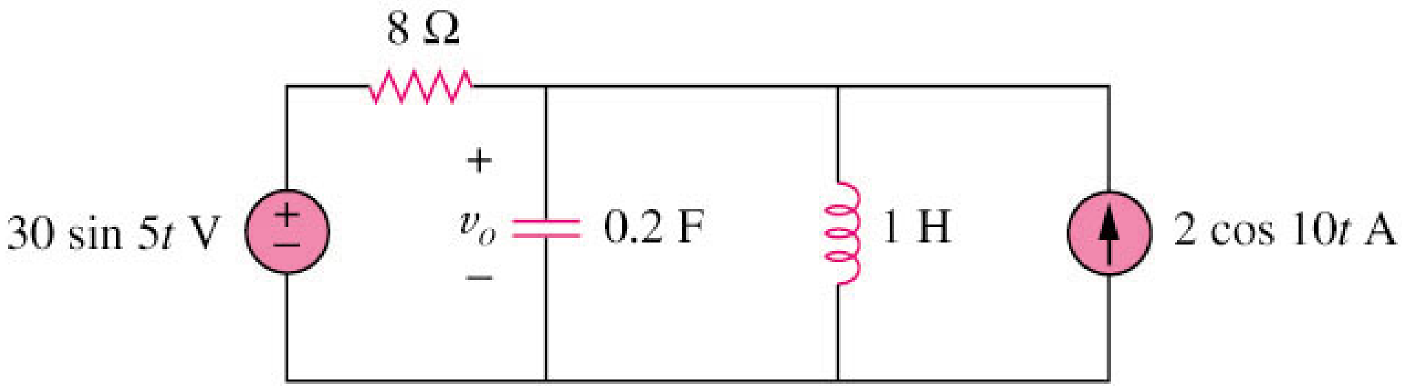
When a circuit has sources operating at different frequencies,

- The separate phasor circuit for each frequency must be solved independently, and
- The total response is the sum of time-domain responses of all the individual phasor circuits.

Superposition Theorem

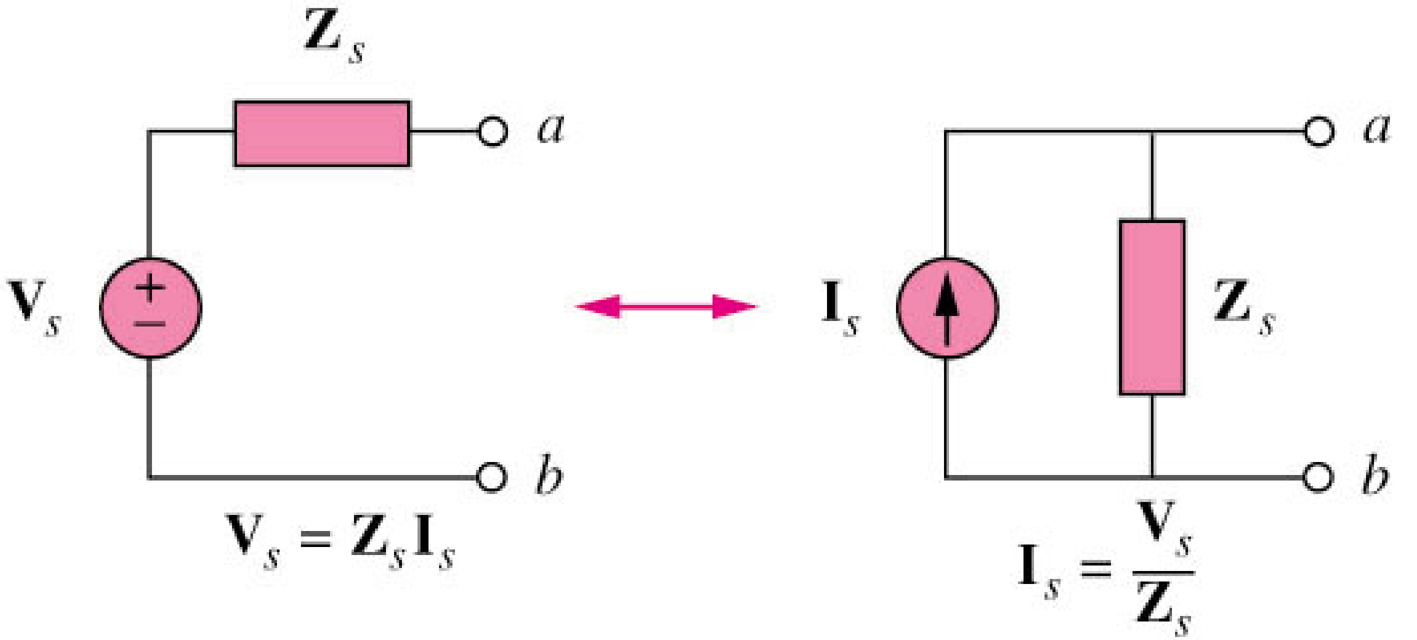
Example 3

Calculate v_o in the circuit of figure shown below using the superposition theorem.



V_o = **4.631 sin(5t – 81.12°) + 1.051 cos(10t – 86.24°) V**

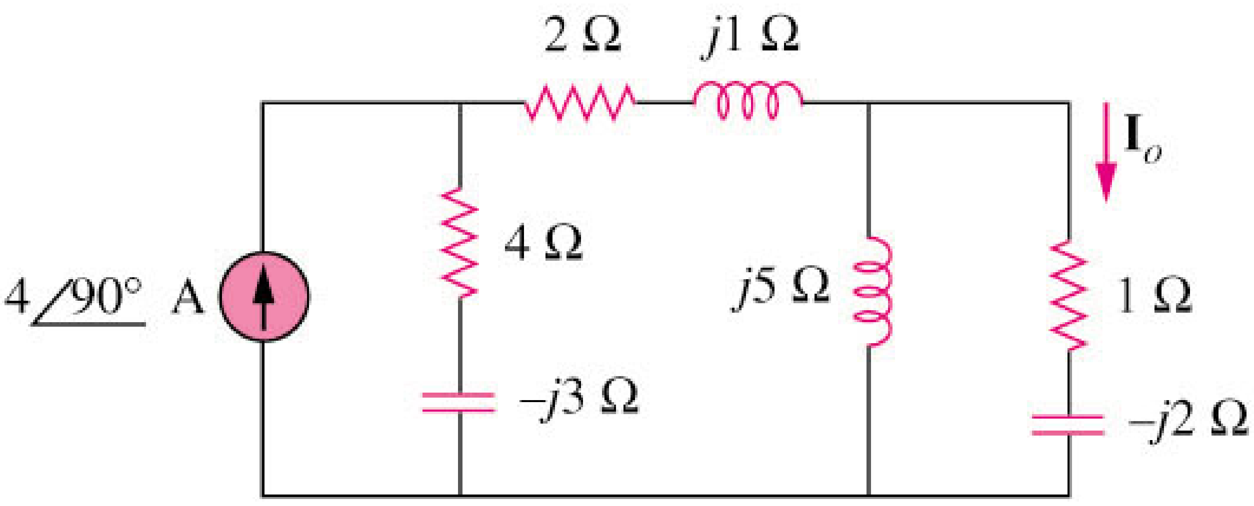
Source Transformation



Source Transformation

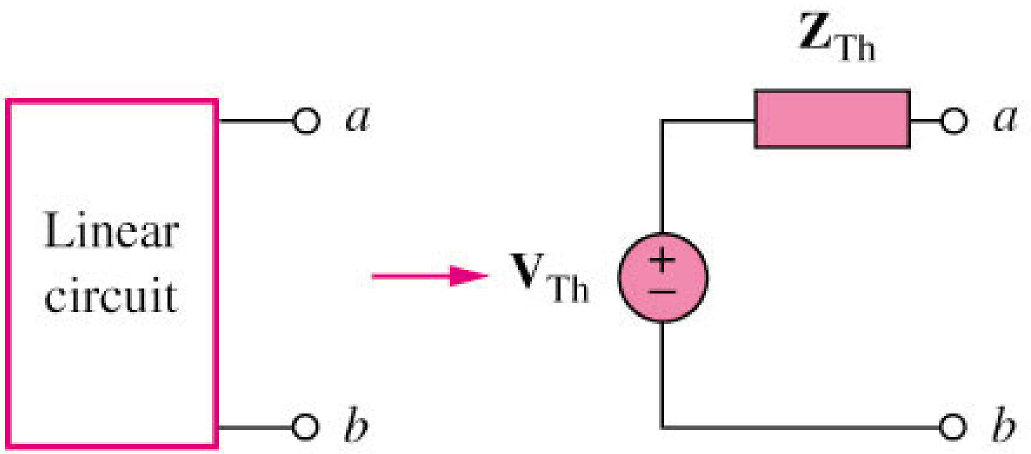
Example 4

Find I_o in the circuit of figure below using the concept of source transformation.

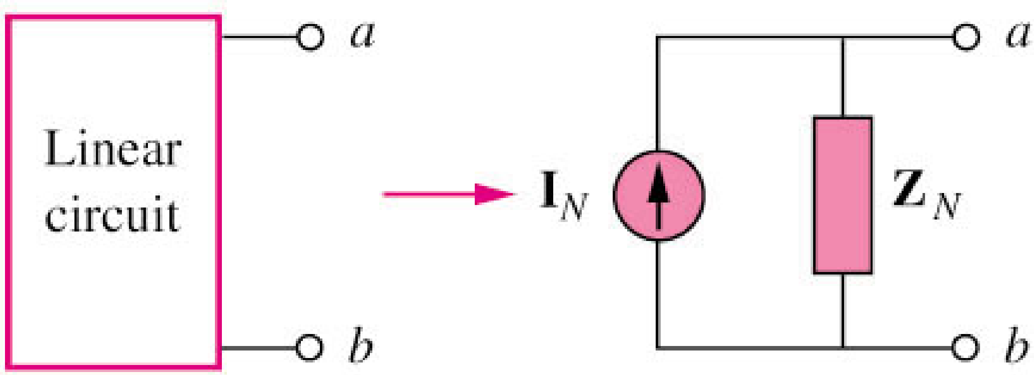


I_o = **3.288 /_99.46^o A**

Thevenin and Norton Equivalent Circuits



Thevenin transform

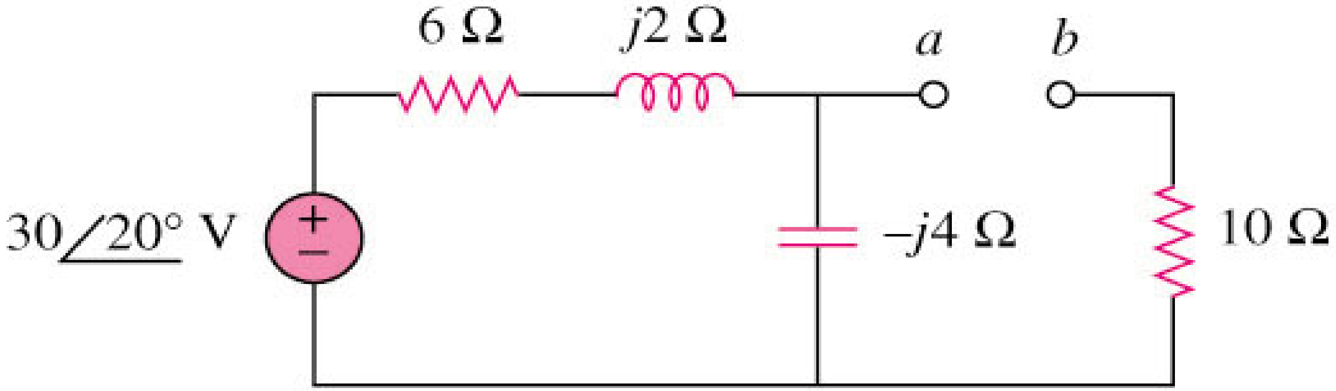


Norton transform

Thevenin and Norton Equivalent Circuits

Example 5

Find the Thevenin equivalent at terminals a–b of the circuit below.



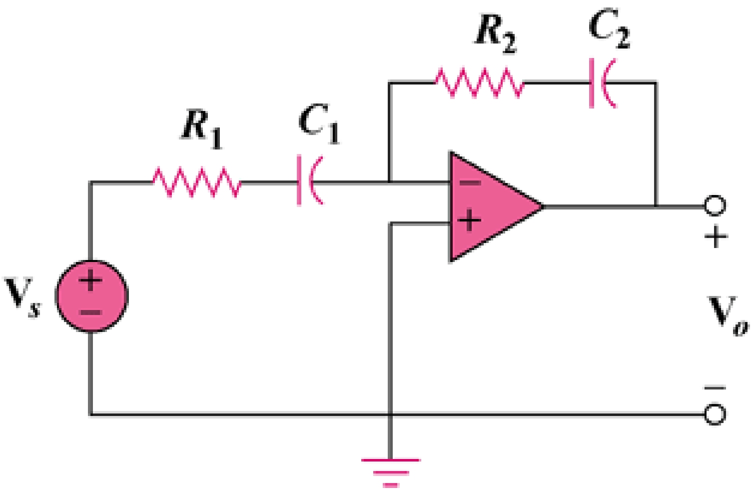
V_{TH} = **18.97 ∠-51.57° V**

Z_{th} =**12.4 – j3.2 Ω**

Op Amp AC Circuit

Example 6

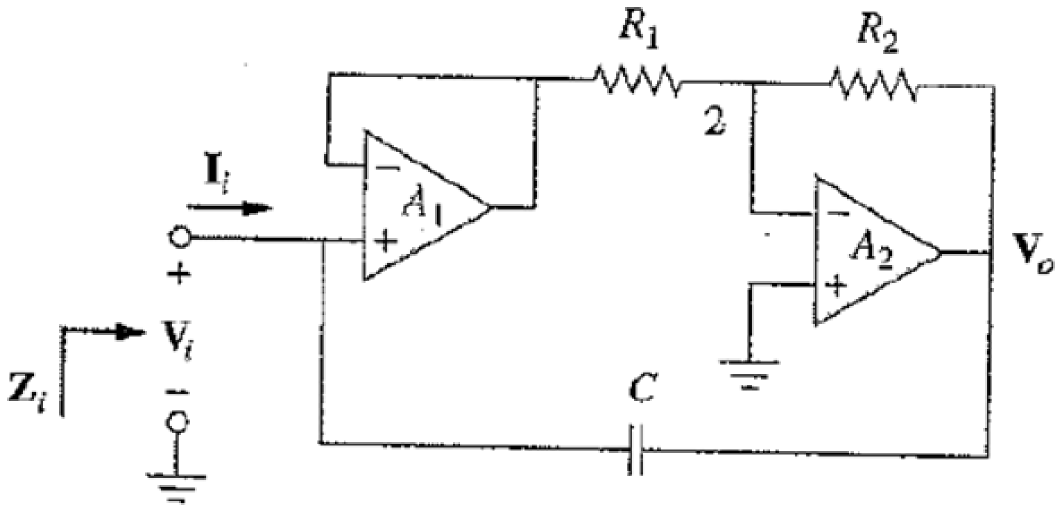
Evaluate the voltage gain $\mathbf{A}_v = \mathbf{V}_o/\mathbf{V}_s$ in the op amp circuit. Find $\mathbf{A}_v$ at $\omega = 0$, $\omega \to \infty$, $\omega = 1/R_1C_1$, and $\omega = 1/R_2C_2$.



Application

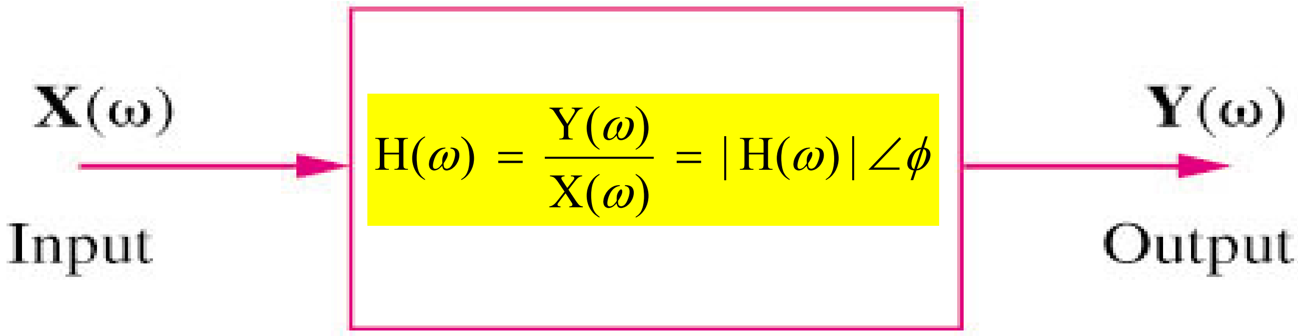
Capacitance Multiplier

Find the equivalent impedance Z_i



Transfer Function H(ω)

- Transfer function (Frequency Response)

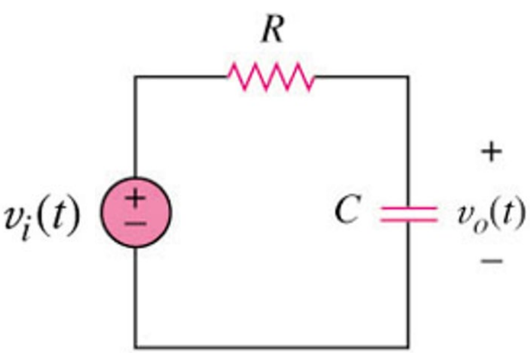


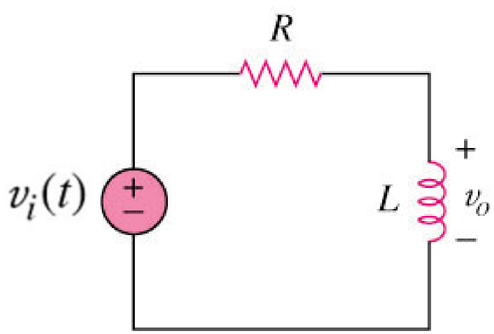
$$H(\omega) = \text{Voltage gain} = \frac{V_o(\omega)}{V_i(\omega)}$$

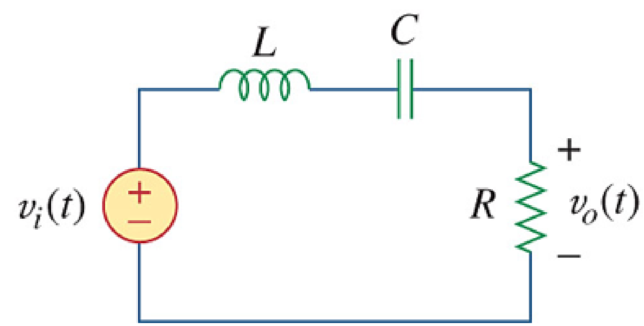
Transfer Function

Example

For the RLC circuits below, obtain the transfer functions V_o/V_i and its frequency response. Let $v_i = v_m \cos\omega t$.







$$H(\omega) = \frac{1}{1 + j\omega RC}$$

$$H(\omega) = \frac{j\omega L}{R + j\omega L}$$

$$H(\omega) = \frac{j\omega RC}{(1 - \omega^2 LC) + j\omega RC}$$